
ABILAKSHAN VADIVEL

SURVEYOR | GIS ANALYST

Professional Interests: Surveying | Hydrographic Survey |
Drone Survey | GIS Analysis | Web GIS

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Jaffna, Sri Lanka

*B.Sc. (Hons) in Surveying Sciences,
specializing in GIS*

PROFESSIONAL SUMMARY

Motivated Surveyor and GIS Analyst with a B.Sc. (Hons) in Surveying Sciences, specializing in GIS, and practical experience in land, topographic, construction, drone, and hydrographic surveying. Skilled in GNSS-based field operations, spatial data preparation, CAD-to-GIS conversion, land information system development, and GIS-based mapping workflows using both Esri and open-source platforms. Able to contribute effectively to field surveying, geospatial data management, mapping, and project support through accurate analysis, reporting, and survey documentation.

TECHNICAL PROFILE

Surveying & GIS Skills:

Boundary Surveying | Topographic Surveying | Engineering Surveying | GNSS-based Field Data Collection | CAD-to-GIS Conversion | Spatial Data Cleaning & Digitizing | Spatial Analysis | Thematic Mapping | Land Information Systems | Web GIS Layer Preparation | WMS/WFS Publishing

Software:

ArcGIS Pro | ArcGIS Online | ArcGIS Experience Builder | QGIS | PostgreSQL | PostGIS | GeoServer | Leaflet | React | MapLibre | AutoCAD | Civil 3D | CloudCompare | Pix4D Mapper | Google Earth Engine | VS Code | MS Office

Instrument Exposure:

GNSS/GPS Receivers | Total Station | Auto Level | Theodolite | Drone Platforms | Terrestrial Laser Scanner (TLS) | Ground Penetrating Radar (GPR)

WORK EXPERIENCE

Surveyor | Riyan Private Limited | Maldives

May 2026 - Present

- Executed drone survey operations using DJI Mavic 3 Enterprise with base station setup and Ground Control Points (GCPs) to ensure accurate aerial data capture.
- Processed and quality-checked UAV outputs in Pix4D Mapper, producing Ortho mosaics and point clouds for survey, mapping, and project documentation purposes.
- Conduct land, topographic, and construction surveys to support site setting-out, verification, progress monitoring, and compliance with project requirements.
- Prepare survey drawings, maps, records, and GIS-based spatial analyses while managing geospatial data for project reporting and decision support.

- Execute drone and hydrographic survey operations, including field data acquisition, processing, quality checks, and integration with project survey outputs.

Planning Intern (Surveyor & GIS Analyst) | Riyan Private Limited
| Maldives

Dec 2025 - March 2026

Land Information System (LIS) using Esri Products (ArcGIS Pro, ArcGIS Online)

- Developed LIS datasets in ArcGIS Pro through CAD-to-GIS conversion, land-use feature creation, and preparation of interactive mapping outputs.
- Published Web Maps, Web Scenes, Experience Builder views, and 3D-ready outputs through ArcGIS Online for land-use planning and visualization.

Land Information System (LIS) using Open-Source GIS & Spatial Database (QGIS, PostgreSQL/PostGIS, GeoServer)

- Transitioned LIS workflows to QGIS, PostgreSQL/PostGIS, and GeoServer while maintaining consistent data structure, layer organization, and project outputs.
- Prepared database-connected layers, supported WMS/WFS publishing, and developed basic Leaflet-based Web GIS functions for portal-ready spatial data access.

Land Demarcation & GNSS Field Support

- Conducted site reconnaissance, checked approved plans, and supported GNSS-based boundary verification and block demarcation activities in the field.
- Maintained field notes, progress records, and completion documentation using reference photographs and drone imagery.

Coastal / Sediment Analysis

- Performed coastal sediment sample preparation and sieve analysis for beach nourishment and coastal protection studies.
- Maintained laboratory records and supporting documentation for sand material suitability assessment.

GIS Outputs, Heat Maps, Harbour Digitization & Drone Exposure

- Produced Resilient Schools heat maps, school sketches, and time-based GIS outputs by checking point alignment and revising layouts based on project requirements.
- Supported harbour digitization, GIS feature editing, and review of Pix4D outputs including orthomosaics, point clouds, DTM, and DSM deliverables.

Junior Surveyor | North Axis Surveys (Pvt) Ltd | Sri Lanka

Oct 2022 - Dec 2025

- Conducted land, boundary, topographic, and control surveys for multiple projects, including survey control establishment, traversing, leveling, and field data collection.

- Operated Total Stations, Sokkia Auto Level, and GNSS equipment, and prepared survey drawings and reports using AutoCAD and GIS software to support land demarcation, site layout, and construction survey activities.

GeoAI for Disaster Resilience - Undergraduate Fellowship | ADRIMP

Feb 2026 - May 2026

- Applied GeoAI, spatial analytics, and risk-modeling concepts to support disaster resilience and risk-reduction planning.
- Worked with geospatial datasets, hazard mapping approaches, and decision-support frameworks for emergency preparedness.
- Strengthened knowledge in resilience-oriented geospatial analysis and GeoAI-based spatial interpretation.

Management Assistant (IT) - Internship | Institute of Medical Sciences, Manipal

Mar 2021 - Aug 2021

- Provided IT support and assisted with database-related tasks to help maintain smooth organizational operations.

SURVEYING PROJECTS

Topographic Survey & Control Establishment | North Axis (Pvt) Ltd

- Participated in a professional topographic survey for a water tank and sump project in Meesalai, Jaffna.
- Established three primary control points and collected field data for contour plan preparation.

Engineering Survey for Reservoir Construction | Academic Field Project

- Conducted engineering and topographic survey tasks for a proposed reservoir and supported access-road design using field survey data.

Survey Camp | Karagasthalawa Maha Vidyalaya, Belihuloya

- Completed topographic and boundary surveys, established control points using static GPS, and prepared site plans for future development.

Advanced Surveying Technology Workshops

- Gained exposure to drones (LiDAR and multispectral), Terrestrial Laser Scanning (TLS), and Ground Penetrating Radar (GPR) through technical workshop sessions.

GIS PROJECTS

Land Information System / Interactive Web GIS Project

- Prepared and organized shapefiles, satellite imagery, drone imagery, and CAD data within a common GIS workflow.

- Contributed to layer symbology, labels, attribute tables, pop-ups, web-layer publishing, and interactive map configuration.
- Supported LIS implementation using ArcGIS-based workflows and open-source tools including QGIS, PostgreSQL/PostGIS, GeoServer, Leaflet, React, and MapLibre.

Landslide Vulnerability Mapping

- Produced a landslide vulnerability map for Matale District using GIS-based spatial analysis and the Analytical Hierarchy Process (AHP).

DEM Comparison and Geostatistical Analysis

- Generated and compared DEMs derived from digitized contours with SRTM and ASTER data and completed geostatistical analysis exercises in R.

Final Year Undergraduate Research

- Completed final-year undergraduate research on GeoAI-based spatial feature detection using geotagged data and multi-temporal Sentinel-2 imagery.
- Applied DeepLabV3+ modelling and achieved 73% accuracy in the completed final model results.

Additional GIS Work

- Created an interactive StoryMap to visualize the impacts of the 2019-2020 Australian bushfires.
- Developed a dynamic dashboard for real-time weather visualization and monitoring.

EDUCATION

B.Sc. (Hons) in Surveying Sciences, specializing in GIS | Sabaragamuwa University of Sri Lanka

- Completed academic requirements | GPA: 3.5 / 4.00 | Faculty of Geomatics

Full Stack Web Development | Uki Coding School

Jun 2020 - Nov 2020

- Online Cohort

- Covered ReactJS, NodeJS, and MongoDB fundamentals.

G.C.E. Advanced Level - Physical Science | St.

2019

John's College, Jaffna

- Results: 2B, 1C

PROFESSIONAL MEMBERSHIPS

- Royal Institution of Chartered Surveyors (RICS) - Professional Number(Student): 0994438
- Geo Mappers
- ISPRS Student Consortium

CERTIFICATES

- Spatial Data Science: The New Frontier in Analytics - MOOC
- Creating an ArcGIS StoryMaps Briefing

- Make an Impact with Modern Geo Apps - MOOC
- Going Places with Spatial Analysis - MOOC
- GIS for Climate Action - MOOC
- Cartography - MOOC
- Telling Stories with GIS Maps
- Introduction to Regression Analysis Using ArcGIS Pro
- Building a Regression Model Using ArcGIS Pro
- Python for Beginners
- Foundations of Project Management

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

- Inter-University Badminton Team Vice-Captain (2025), Member (2022-2025)
- Inter-University Table Tennis Runners-Up (2022), Member (2022-2025)
- Tamil Cultural Society, SUSL - Program Coordinator (2022), Media Coordinator (2023), Faculty of Geomatics Representative (2024)
- School Badminton Team Captain (2017), Member (2012-2018)
- School Volleyball Team Captain (2018), Member (2017-2018)
- School Cricket Team Member (2011-2018)
- School Prefect (2016-2018)
- Best Cricket Team in Northern Province (2017)
- All-Island School Cricket Under-13 Runners-Up (2013)

REFERENCES

Prof. RMKGSPB Koswatte

Professor

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